



Experiment-7

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Semester: 5th

Subject Name: Computer Networks

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Date: 09/09/2026

Subject Code: 24CSH-337

1. Aim/Overview of the practical:

To implement and configure a **Cisco router as a DHCP server** that can provide IP addresses to multiple VLANs using **Inter-VLAN Routing and Router-on-a-Stick** in Cisco Packet Tracer.

2. Software/Tools Required:

- Cisco Packet Tracer
- Windows Operating System

3. Theory:

A) VLAN (Virtual Local Area Network)

VLAN is a logical network created within a physical switch. It divides devices into separate broadcast domains.

Example:

- VLAN 10 → STUDENTS
- VLAN 20 → FACULTY

VLANs provide network segmentation, security, and better network management.

B) Inter-VLAN Routing

Different VLANs are separate Layer 2 networks. Communication between them requires a Layer 3 device such as a router. This process is called Inter-VLAN Routing.

C) Router-on-a-Stick

Router-on-a-Stick uses one physical router interface with multiple logical subinterfaces to perform Inter-VLAN Routing.

Example:

- G0/0.10 → VLAN 10
- G0/0.20 → VLAN 20

D) VLAN Tagging

VLAN tagging identifies the VLAN to which an Ethernet frame belongs when it travels through a trunk. IEEE 802.1Q is commonly used for VLAN tagging.

E) DHCP

DHCP (Dynamic Host Configuration Protocol) automatically provides devices with IP address, subnet mask, default gateway, and other network information.

F) DHCP Server and DHCP Pools

A Cisco router can act as a DHCP server. Separate DHCP pools can be created for different VLANs.

Example:

- VLAN 10 → 192.168.10.x
- VLAN 20 → 192.168.20.x

The router automatically assigns IP addresses to PCs according to their VLAN.

G) DHCP DORA Process

DHCP works through four steps:

- D – Discover
- O – Offer
- R – Request
- A – Acknowledgment

4. Steps for Experiment/Practical:

A) Create the Network Topology

Procedure / Steps:

- 1) Open Cisco Packet Tracer and place 1 Cisco 2911 Router, 1 Cisco 2960 Switch, and 4 PCs. Rename them R1, S1, PC1, PC2, PC3, and PC4.
- 2) Connect:
 - PC1 → S1 Fa0/1
 - PC2 → S1 Fa0/2
 - PC3 → S1 Fa0/3
 - PC4 → S1 Fa0/4



3) S1 Fa0/24 → R1 G0/0

- On S1 CLI, create VLANs:
- enable
- configure terminal
- vlan 10
- name STUDENTS
- exit
- vlan 20
- name FACULTY
- exit

4) Assign PC1 and PC2 to VLAN 10:

- interface range fastEthernet 0/1-2
- switchport mode access
- switchport access vlan 10
- exit

5) Assign PC3 and PC4 to VLAN 20:

- interface range fastEthernet 0/3-4
- switchport mode access
- switchport access vlan 20
- exit

6) Verify VLANs:

- show vlan brief

7) Configure Fa0/24 as a trunk:

- interface fastEthernet 0/24
- switchport mode trunk
- exit

8) Verify the trunk:

- show interfaces trunk

9) On R1 CLI, enable the physical interface:

- enable
- configure terminal
- interface gigabitEthernet 0/0
- no shutdown
- exit

10) Configure the VLAN 10 subinterface:

- interface gigabitEthernet 0/0.10
- encapsulation dot1Q 10
- ip address 192.168.10.1 255.255.255.0
- exit

11) Configure the VLAN 20 subinterface:

- interface gigabitEthernet 0/0.20
- encapsulation dot1Q 20
- ip address 192.168.20.1 255.255.255.0

- exit

12) Configure DHCP excluded addresses:

- ip dhcp excluded-address 192.168.10.1
- ip dhcp excluded-address 192.168.20.1

13) Create the VLAN 10 DHCP pool:

- ip dhcp pool VLAN10_POOL
- network 192.168.10.0 255.255.255.0
- default-router 192.168.10.1
- dns-server 8.8.8.8
- exit

14) Create the VLAN 20 DHCP pool:

- ip dhcp pool VLAN20_POOL
- network 192.168.20.0 255.255.255.0
- default-router 192.168.20.1
- dns-server 8.8.8.8
- exit

15) On PC1, PC2, PC3, and PC4, go to Desktop → IP Configuration → DHCP to obtain IP addresses automatically.

16) Verify DHCP assignments:

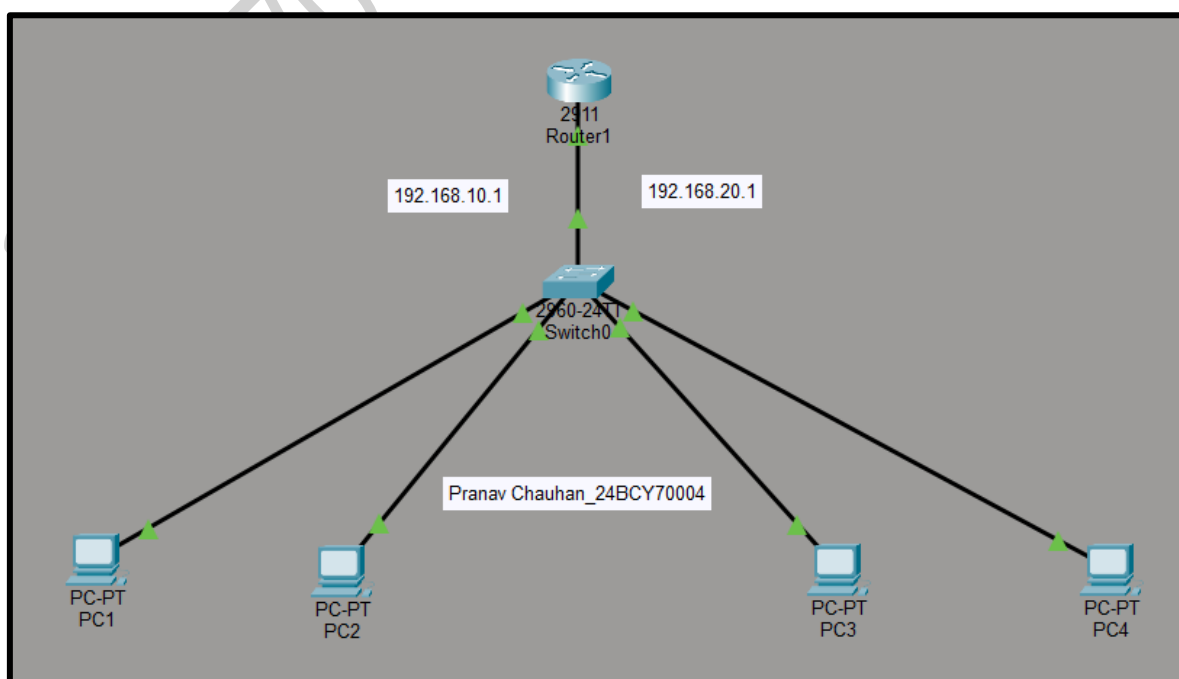
- show ip dhcp binding

17) Verify router interfaces:

- show ip interface brief

18) Test connectivity between PCs and VLAN gateways using the ping command.

5. Output:



6. Result

The router was successfully configured as a **DHCP server for multiple VLANs**. VLAN 10 and VLAN 20 were configured with separate DHCP pools, and the PCs automatically received IP addresses from their respective VLAN networks. Inter-VLAN communication was successfully verified using the router.

7. Learning Outcomes

- Understood the concept of VLANs and Inter-VLAN Routing.
- Learned to configure multiple VLANs on a switch.
- Learned to configure trunk and access ports.
- Understood Router-on-a-Stick and router subinterfaces.
- Learned to configure a router as a DHCP server.
- Created separate DHCP pools for different VLANs.
- Learned to verify VLAN, trunk, routing, and DHCP configurations.